

MILES SCHEETZ

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Work Experience

Mechanical Engineering Co-op

January 2026 to present

Hyperion Solutions, LLC – Memphis, TN

- Computed stresses on elevator components with hand calculations and FEA to verify ASME A17.1 compliance
- Created SolidWorks models and production drawings for 4 new product assemblies, including BOMs, GD&T, material specifications, and manufacturing notes
- Prototyped spring-loaded encoder wheel assembly, iterating 3D printed concepts, testing different tension mechanisms, and varying tire material to minimize slippage
- Revised 100+ pages of traction machine manuals to clarify installation and maintenance instructions

Manufacturing Engineering Co-op

January 2025 to January 2026

Entegris, Inc. – Chaska, MN

- Qualified 80+ products on secondary machines to increase routing flexibility and reduce capacity bottlenecks
- Drove 15 Management of Change (MOC) projects to improve machining processes
- Initiated, qualified, and implemented 130+ ECRs/ECOs for manufacturing process changes
- Programmed Rockwell PLC and HMI for controlling exhaust fan system with ladder logic

Rotary Press Operator

October 2021 to January 2025

Innovize, LLC – St. Paul, MN

- 3+ years of hands-on manufacturing experience setting up rotary presses and diagnosing mechanical issues
- Coordinated assembly line with 1–5 technicians during medical device production runs
- Spearheaded lean project for Medtronic Oval Tape production, improving process reliability by 230% by implementing system to detect and alarm when input material is low

Projects & Leadership

Bethel University Rocket Club

- Built fiberglass rocket for 2025 NASA Space Grant Consortium Midwest Competition; placed 1st out of 13 teams
- Conducted FEA on custom-made rocket components with SolidWorks
- Designed and constructed an avionics bay using Arduino to track and store flight data
- Developed and taught 5-week training curriculum on rocketry basics to 10+ new recruits
- Mentored 10+ students on academic performance and career development
- Achieved Level 2 High Power Rocketry certification under NAR

Railgun Hobby Project

- Calculated projectile kinematics and capacitor discharge behavior in Mathematica to determine design criteria
- Modeled railgun body in SolidWorks using copper rails, 3D-printed components, and neodymium magnets
- Designed unique spring-loaded armature to enhance rail contact and reduce wear
- Increased hand-tool fabrication accuracy to within $\pm 0.005''$ by creating 3D-printed drilling and alignment fixtures

Education

B.S. Mechanical Engineering

Expected graduation May 2027

Bethel University – St. Paul, MN

August 2023 to Present

- 65 credits, 3.56 GPA, Dean's List Spring and Fall 2024
- Relevant Coursework: Engineering Materials and Manufacturing, Electronics, Mechanics, and Fluid Dynamics
- Member of Rocket Club and Sigma Pi Sigma honor society

Inver Hills Community College – St. Paul, MN

September 2022 to May 2023

- 20 credits, 3.75 GPA

Skills & Certifications

- SolidWorks (Certified SolidWorks Professional) and Lean Manufacturing (MFRALL Certified Lean Practitioner)
- SolidWorks Simulation FEA, SolidWorks 2D Drawings, 3D-Printing, Python, Arduino, MS Office, Industrial Machining, Medical Device QMS, GD&T, Control Systems, PLC Programming, SAP ERP